Elliot Tower
Independent Researcher
elliot@elliottower.ai

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Editorial Office
*Genetic Epidemiology*
Wiley

Dear Editors,

Please consider our manuscript, "Abundance or Activity? Mechanism-Dependent Validity of *Cis*-pQTL Mendelian Randomization," for publication as an Original Research Article in *Genetic Epidemiology.*

This study addresses a methodological question central to genetic epidemiology: when does a *cis*-pQTL instrument provide a valid proxy for a pharmacologic intervention? Current evaluations of pQTL-based Mendelian randomization for drug-target validation pool all drug mechanisms and report composite performance. This manuscript shows that such pooling obscures a mechanism-dependent boundary of validity: *cis*-pQTL MR is informative when the drug perturbs protein abundance, but structurally uninformative when the drug acts by blocking protein function without changing measured protein level.

The analysis is a pre-registered, outcome-blind evaluation of a frozen zero-parameter classifier across 195 Phase III drug-target pairs. The main contribution is not simply a benchmark, but an instrument-validity framework that decomposes composite pQTL performance into interpretable mechanistic strata and helps explain why recent pQTL benchmarks have reported weaker enrichment than GWAS-based approaches.

The manuscript is well aligned with *Genetic Epidemiology* because it focuses on the validity, interpretation, and limitations of genetic instruments rather than on a single therapeutic area. All data sources are public, the full protocol history was SHA-256 hashed before execution, and code and classification tables are available in a public repository. The manuscript is not under consideration elsewhere.

Sincerely,

Elliot Tower
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